

Case Study: Boeing 737 MAX & The MCAS Software Failure

Milestone 1. Failure Reconstruction -

Activity 1 — What Actually Happened?

- **System Purpose:** Provide a fuel-efficient variant of the 737 (737 MAX) that handled identically to legacy 737 Next Generation (NG) models, eliminating the requirement for mandatory, costly flight simulator training for airline crews.
- **Primary Stakeholders:** Boeing (Manufacturer), Federal Aviation Administration (FAA - Regulator), Airlines (Customers), Flight Crews/Pilots, Commercial Passengers.
- **Immediate Failure:** Erroneous Angle of Attack (AoA) sensor readings triggered the MCAS software to repeatedly command nose-down stabilizer trim without pilot input or awareness.
- **System Consequence & Impact:** Loss of two passenger aircraft (Lion Air 610 in October 2018; Ethiopian Airlines 302 in March 2019), 346 fatalities, worldwide grounding of the 737 MAX fleet for 20 months, and billions of dollars in losses and legal penalties for Boeing.

Deliverable D1 : Failure Timeline

Phase / Date	Event Description
2011 Business Context	Boeing Launches 737 MAX To compete with the Airbus A320neo without undergoing a lengthy recertification process, Boeing fits larger engines onto the existing airframe, altering aerodynamics and creating extra lift under high thrust.
2012 – 2017 Development & Scope Creep	MCAS Implementation Engineers implement MCAS to automatically adjust trim and prevent high-AoA stalls. Software authority is quietly increased from 0.6° to 2.5° of stabilizer movement without updating safety assessments.

2017 Certification & Omission	Regulatory Approval Boeing intentionally omits MCAS from pilot manuals and training materials to preserve "Level D" simulator-free certification status for airlines.
Oct 29, 2018 First Failure Event	Lion Air Flight 610 Crash A single faulty Angle of Attack (AoA) sensor feeds false "nose-high" data to MCAS. Software repeatedly trims the nose down; pilots fight the system without knowing MCAS exists (189 fatalities).
Nov 2018 Mitigation	Initial Directive Issued Boeing issues an Emergency Airworthiness Directive explaining runaway trim recovery. The core single-sensor architectural flaw remains unpatched.
Mar 10, 2019 Second Failure Event	Ethiopian Airlines Flight 302 Crash An AoA sensor fails after takeoff. Pilots follow memory items but suffer "aerodynamic lock" on the manual trim wheel, losing control (157 fatalities).
Mar 13, 2019 Operational Consequence	Global Fleet Grounding Global aviation authorities (EASA, CAAC, FAA) ordered the worldwide grounding of the entire Boeing 737 MAX fleet.

Milestone 2. Root-Cause Analysis -

Activity 2 — Why did it happen?

5-Why Analysis:

- **C1 — Requirements Failure:** Missing **safety and data-validation requirements for MCAS**. Requirements failed to mandate dual-sensor cross-checking (voting logic) (totally relied on single sensor on the nose), omitted Single-Point-of-Failure (SPOF) protection,

and lacked bounding limits on how much stabilizer trim authority the software could command. Resulting , under the thrust and steep turn leading make the engine pods creating extra lift

- **C2 — Assumption Failure** : Flawed engineering assumptions regarding sensor integrity and pilot workload. Engineers incorrectly assumed a single AoA sensor would not deliver corrupt data and assumed flight crews could diagnose runaway trim within 3 seconds and manually turn the trim wheel under extreme aerodynamic force.
- **C3 — Implementation Failure**: Software execution flaws where source code lacks boundary validation , Mathematical and structural logic flaws where control algorithms lack bounding limits or termination criteria , data-type errors, state-management errors.
- **C4 — Testing / Verification Failure**: Missing test cases, inadequate test coverage, poor integration testing, insufficient regression testing.
- **C5 — Configuration / Change-Management Failure**: Wrong version, uncontrolled changes, inadequate approval, configuration mismatch.

Deliverable D2: Root-Cause Map

Cause Level	Detailed Cause Description	Cause Category	Evidence / Source Material
Immediate Cause	MCAS repeatedly applied uncommanded 2.5° nose-down stabilizer trim, overpowering crew control inputs.	C4 Implementation Failure C8 Human Factor	Flight Data Recorders (FDR) showing periodic 10-second MCAS activations.
Contributing Cause 1	The system relied on a single AoA sensor with no cross-checking against the second sensor on the aircraft	C3 Design Failure	Boeing MCAS Architectural Diagrams; NTSB & Indonesian KNKT reports.
Root Cause 1	System requirements lacked basic fault-tolerance, single-point-of-failure (SPOF) protection, and hard authority limits.	C1 Requirements Failure	Initial System Safety Assessment (SSA) and software requirements specification.

Root Cause 2	Corporate pressure to avoid simulator training led to deliberate omission of MCAS from flight manuals and training materials.	C9 Mgmt Failure C10 Ethics	Internal Boeing communications ("Jedi mind-trick" emails uncovered by US House Inquiry).
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Milestone 3: SDLC Origin/Detection Analysis

Activity 3 - Where did engineer failure

Technical Design Failures

- **Single-Sensor Dependency:** MCAS relied on input from only one Angle of Attack (AoA) sensor, ignoring standard aerospace safety principles of data redundancy and cross-checking
- **The "Pilot as Backup" Fallacy:** Engineers assumed human pilots would instantly diagnose and override erratic software behavior, despite the crew having no prior knowledge of MCAS's existence

Deliverable D3: SDLC Failure Map & Origin–Detection Matrix

SDLC PHASE	ORIGIN OF DEFECT	DETECTION / ESCAPE
1. Requirements	C1 Allowed Single Point of Failure (SPOF) and unlimited activation cycles.	MISSED No safety requirement specified for dual-sensor voting.
2. Design & Architecture	C3 Increased authority to 2.5° without system re-evaluation.	MISSED Hazard classification was not appropriately upgraded.

3. Implementation	C4 Software loop reset on pilot switch action (infinite loop).	MISSED Code review focused strictly on functional compliance.
4. Testing		[DETECTION FAILURE] C5 Simulators failed to model aerodynamic lock or high-stress pilot reaction.
5. Release & Operations		[DETECTION FAILURE] C7/C8 Manuals concealed MCAS; UI lacked alerts; training omitted.

Milestone 4: Engineering & Ethics Analysis (M4) –

1. **Principle of Fail-Safe Operation:** Critical automated systems must default to a safe state upon detecting invalid inputs or anomalous hardware behavior.
2. **Principle of Redundancy & Fault Tolerance:** No single point of failure (SPOF) should be capable of compromising safety-critical operation.
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Deliverable D4: Engineering Responsibility & Ethics Map

Responsible Actor	Duty & Responsibility	Missed Action / Violation	Consequence & Impact
Boeing Executive Management	Maintain safety as paramount; foster a culture where	Prioritized cost reduction and launch schedule	Compromised safety culture; built corporate incentives around avoiding

	technical concerns are welcomed over financial targets.	over system safety (C9). Applied pressure to avoid simulator training.	safety reviews.
Lead Systems & Software Engineers	Adhere to professional ethics (e.g., NSPE Code); ensure fault tolerance in critical architecture.	Accepted single-sensor inputs (C3); failed to escalate risks associated with expanding MCAS trim authority (C10).	Released a system vulnerable to single point of failure (SPOF).
Flight Crew Operations / Technical Writers	Provide accurate, complete operational manuals and training documentation to operators.	Concurrently agreed to remove references to MCAS from the Flight Crew Operations Manual (FCOM) (C10, C8).	Deprived pilots of informed awareness and critical emergency diagnostic information.

Milestone 5 : Failure Prevention -

Requirements Stage-

Mandatory Dual-Sensor Input Validation

- **Classification:** Preventive
- **Failure Addressed:** Eliminates the single point of failure by requiring simultaneous data feeds from both Angle-of-Attack (AoA) sensors instead of relying on just one.

AoA Disagree Alert Requirement

- **Classification:** Detective
- **Failure Addressed:** Addresses the lack of situational awareness by making the cockpit "Disagree Light" mandatory. This alerts pilots immediately when the left and right sensors drift apart.

Design Stage-

Software Authority and Repetition Limits

- **Classification:** Preventive
- **Failure Addressed:** Stops MCAS from continuously compounding nose-down trim commands. It restricts the system to a single activation cycle per high-AoA event. This ensures the total trim change remains small enough for manual physical counteraction.

Yoke Force Pilot Override Priority

- **Classification:** Preventive
- **Failure Addressed:** Prevents automated systems from locking out the pilot. Applying manual force to the control column immediately cuts off and inhibits automated trim inputs.

Testing Stage

Realistic Human-in-the-Loop Stress Testing

- **Classification:** Detective
- **Failure Addressed:** Corrects flawed safety assumptions that expected pilots to diagnose and counter runaway trim within an unrealistic

Deliverable D5: Failure Prevention Plan

SDLC Stage & Type	Control Name	Specific Engineering Action

Design <i>(Preventive)</i>	Hard Trim Authority Bounding Limits	Enforce an absolute software cap on nose-down trim magnitude per event. Ensure control column pull always maintains mechanical override priority over automated commands.
Testing <i>(Detective)</i>	High-Load Human-in-the-Loop Simulation	Evaluate crew recovery procedures under dynamic, high-speed flight regimes. Test physical trim wheel forces under simulated aerodynamic lock conditions.
Release & Deployment <i>(Preventive)</i>	Mandatory Documentation & Transparency	Ban the omission of flight control software behaviors from flight crew manuals. Mandate formal safety re-assessments for any modifications altering flight characteristics.